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Recombination of positive ions and electrons in a dense neutral gas

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Abstract. A theory is presented of the recombination of electrons and positive ions in the presence of neutral atoms which act as third bodies, explicit account being taken of the effect of radiative transitions. Comparison is made with the theories of Thomson and of Pitaevskii. Detailed calculations are carried out on recombination in helium gas. It is found that the rate coefficient α is very great when the temperature θ is low and the number density $N(\text{He})$ is high: for example, that α for He^+ is $5 \times 10^{-8} \text{ cm}^3 \text{ sec}^{-1}$ and for Na^+ is $2 \times 10^{-8} \text{ cm}^3 \text{ sec}^{-1}$ when θ is 250 °K and $N(\text{He})$ is $3 \times 10^{19} \text{ cm}^{-3}$.

1. Introduction

Thomson (1924) postulated that a positive and a negative ion recombine if either suffers a collision with a third body when at a distance from the other less than

$$r_0 = 2e^2/3k\theta \quad (1)$$

where θ is the temperature. He hence deduced that ions which pass within the critical distance have a probability

$$\mathcal{P}_i = \frac{4r_0}{3} \left\{ \frac{1}{l_i^+} + \frac{1}{l_i^-} \right\} \quad (2)^\ddagger$$

of recombining, l_i^+ and l_i^- being their mean free paths. In the case of three-body ion-electron recombination he suggested that the corresponding probability is

$$\mathcal{P}_e = \frac{4r_0}{3} \frac{\lambda}{l_e} \quad (3)$$

where l_e is the electron's mean free path and λ is the mean fraction of its kinetic energy which is lost in a collision. We shall take the third body to be an atom X of mass M so that

$$\lambda = 2m/M \quad (4)$$

where m is the electronic mass (cf. Massey and Burhop 1952). Thomson's theory then gives that the recombination coefficient is

$$\alpha_T = \frac{64(3\pi^2)^{1/2}}{81} \frac{e^8 m^{1/2}}{M(k\theta)^{5/2}} N(X) \sigma \quad (5)$$

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‡ A correction which Loeb (1939) made to the original formula for the probability is included and it is assumed that l_i^+ and l_i^- are much greater than r_0 .

$N(X)$ being the number density of the atoms and σ the momentum transfer cross section for electron-atom collisions.

Pitaevskii (1962) has recently presented an elegant solution to the problem. Regarding the recombination process as classical diffusion in energy space and applying the Fokker-Planck equation he found that the recombination coefficient is

$$\alpha_P = \frac{1}{4} \left(\frac{\pi e^4}{k\theta} \right)^{3/2} \mathcal{J}^{-1} \quad (6)$$

with

$$\mathcal{J} = \int_{-\infty}^0 \frac{|E|^{5/2} \exp(E/k\theta)}{(\partial\Delta^2/\partial t)_E} dE \quad (7)$$

in which E is the energy of the electron and $(\partial\Delta^2/\partial t)_E$ is the average of the square of the change per collision of this energy multiplied by the number of collisions per unit time. If $\mathbf{V}_{e1}$ and $\mathbf{V}_{e2}$ are the velocities of the electron relative to the ion before and after a collision and if $\mathbf{V}_{a1}$ and $\mathbf{V}_{a2}$ are the corresponding relative velocities of the atom then

$$\Delta = \frac{1}{2}m(V_{e2}^2 - V_{e1}^2) = m\mathbf{G} \cdot (\mathbf{V}_{e2} - \mathbf{V}_{e1}) \quad (8)$$

where

$$\mathbf{G} = (m+M)^{-1}(M\mathbf{V}_{a1} + m\mathbf{V}_{e1}) \quad (9a)$$

$$= (m+M)^{-1}(M\mathbf{V}_{a2} + m\mathbf{V}_{e2}). \quad (9b)$$

Pitaevskii neglected the second terms on the right of (9a) and (9b) in comparison with the first terms and assumed the scalars V_{e1} and V_{e2} to be the same. Assuming that the differential scattering cross section is independent of the vectors $\mathbf{V}_{e1}$ and $\mathbf{V}_{e2}$ he obtained that

$$\left(\frac{\partial\Delta^2}{\partial t} \right)_E = \frac{128(2)^{1/2}}{3\pi} \frac{m^{1/2}k\theta N(X)\sigma}{M} |E|^{3/2} \quad (10)$$

and hence from (6) and (7) that

$$\alpha_P = \frac{32(2\pi)^{1/2}}{3} \frac{e^6 m^{1/2}}{M(k\theta)^{5/2}} N(X)\sigma. \quad (11)$$

Comparison of (5) and (11) shows that

$$\frac{\alpha_P}{\alpha_T} = \frac{9}{2} \left(\frac{6}{\pi} \right)^{1/2} \simeq 6.2. \quad (12)$$

It would not have been surprising if this ratio had differed much more from unity: thus the approximation to equations (9a) and (9b) used in the derivation of the final expression for α_P makes the energy loss parameter λ , to which α_T is proportional, vanish. The justification for assertion (3) on the probability of recombination is far from obvious. Pitaevskii's general formula (6) is undoubtedly a great advance. However, the use of the Fokker-Planck equation when the energy levels do not form a continuum causes some uncertainty. It is desirable to express the three-body recombination coefficient directly in terms of the rate coefficients of the two-body processes involved and to make explicit allowance for radiative transitions which are in some circumstances important.

We shall assume throughout that the free electrons are not sufficiently abundant to be important as third bodies (cf. Bates, Kingston and McWhirter 1962).

2. General theory

We shall suppose that the electrons e are recombining with ions X^+ in a gas composed of the parent atoms; and we shall suppose that the excited levels through which the electrons pass are hydrogenic and that there is a uniform distribution amongst the degenerate states of a level. Let $\kappa(n, m)$ be the rate coefficient for the inelastic collision process



where n and m are the principal quantum numbers of the initial and final states; let

$$\epsilon(n) = I/k\theta n^2 \quad (14)$$

where I is the ionization potential of hydrogen. Using an approximate formula which Bates and Kingston (1964) obtained by considering the down-flow of electrons in the quasi-equilibrium steady state, we have that the recombination coefficient for



is

$$\alpha_c = \left(\frac{h^2}{2\pi m k \theta} \right)^{3/2} \left\{ \sum_{q=q_0+1}^{\infty} \frac{1}{\mathcal{C}(q)} \right\}^{-1} \quad (16)$$

where q_0 is the principal quantum number of the highest level whose population may rise sufficiently to deplete the population of the continuum appreciably and where

$$\mathcal{C}(q) = N(X) \sum_{s=0}^{\infty} \sum_{t=0}^{q-q_0-1} (1+s+t)(q+s)^2 \kappa(q+s, q-1-t) e^{\epsilon(q+s)} \quad (17)$$

the subscript C on α being a reminder that only collisional processes are taken into account. Formula (16) describes pure collisional recombination satisfactorily provided the higher terms in expansion (17) are not unduly important. Fortunately, this expansion converges rapidly, as will be apparent later. It may be conveniently written as

$$\mathcal{C}(q) = N(X) q^2 \{ \kappa(q, q-1) + 4\Lambda_1 \kappa(q, q-2) + 9\Lambda_2 \kappa(q, q-3) + \dots \} e^{\epsilon(q)} \quad (18)$$

with

$$\begin{aligned} \Lambda_1 &= \frac{1}{2} \left\{ 1 + \frac{(q+1)^2 \kappa(q+1, q-1) e^{\epsilon(q+1)}}{q^2 \kappa(q, q-2) e^{\epsilon(q)}} \right\} \\ \Lambda_2 &= \frac{1}{3} \left\{ 1 + \frac{(q+1)^2 \kappa(q+1, q-2) e^{\epsilon(q+1)} + (q+2)^2 \kappa(q+2, q-1) e^{\epsilon(q+2)}}{q^2 \kappa(q, q-3) e^{\epsilon(q)}} \right\}. \end{aligned} \quad (19)$$

The relation between α_p of (6) and α_c of (16) will now be examined. A considerable contribution to the classical expression (10) for $(\partial \Delta^2 / \partial t)_E$ arises from collisions in which the electron suffers only a minute change in energy, other collisions being relatively infrequent. The value obtained for the average falls when allowance is made for the separation between the energy levels. In the case of an electron in level q we have

$$\left(\frac{\partial \Delta^2}{\partial t} \right)_q = N(X) I^2 \sum_{r=1}^{\infty} \left\{ \frac{1}{q^2} - \frac{1}{(q \pm r)^2} \right\} \kappa(q, q \pm r) \quad (20)$$

where we have defined

$$\kappa(q, q-r) = 0 \quad \text{for } r \geq q. \quad (21)$$

If the energy levels formed a continuum, the probability of an upward transition would

be the same as that of a downward transition on Pitaevskii's approximation. However, because level $q + r$ lies closer to level q than does level $q - r$ the rate coefficient $\kappa(q, q + r)$ may exceed the rate coefficient $\kappa(q, q - r)$ by a considerable factor which makes the use of a formula based on the Fokker-Planck equation dubious. Ignoring this and substituting from (20) into (7) we find that instead of being given by (11) the recombination coefficient is

$$\alpha_P' = \left(\frac{h^2}{2\pi m k \theta} \right)^{3/2} \left\{ \sum_{q=q_0+1}^{\infty} \frac{1}{\mathcal{P}(q)} \right\}^{-1} \quad (22)$$

with

$$\mathcal{P}(q) = \frac{1}{8} N(X) q^4 e^{e(q)} \left[\sum_{r=1}^{\infty} r^2 \left\{ \frac{(2q+r)^2}{(q+r)^4} \kappa(q, q+r) + \frac{(2q-r)^2}{(q-r)^4} \kappa(q, q-r) \right\} \right]. \quad (23)$$

It is seen that (16) and (22) have the same structure. There is also a close correspondence between a term of (18) and the term of (23) for which the modulus of the change in the principal quantum number is the same. In the limit

$$q \rightarrow \infty, \quad \theta \rightarrow 0 \quad (24)$$

it may be seen that

$$\mathcal{C}(q) \rightarrow \mathcal{P}(q); \quad (25)$$

but when q is finite and θ is non-zero

$$\mathcal{C}(q) < \mathcal{P}(q) \quad (26)$$

so that

$$\alpha_C < \alpha_P' (< \alpha_P). \quad (27)$$

Pitaevskii recognized that formula (11) greatly overestimates the recombination coefficient at the higher temperatures and gave as a condition for its validity that

$$\theta \ll \theta_P \quad (28)$$

where

$$\theta_P = \left(\frac{m}{M} \right)^{1/2} \frac{e^2}{k a_0} = \frac{7 \times 10^3}{(M_{\text{Ch}})^{1/2}} \text{ } ^\circ\text{K} \quad (29)$$

the subscript Ch indicating that M is here on the chemical scale.

2.1. Radiative transitions

We write

$$f(q) = \mathcal{C}(q) / \mathcal{A}(q) \quad (30)$$

where $\mathcal{C}(q)$ is as in (17) and where

$$\mathcal{A}(q) = q^2 A(q) e^{e(q)} \quad (31)$$

in which $A(q)$ is the sum of the spontaneous transition probabilities from level q to all lower levels except the ground level (which we exclude because of self-absorption). This ratio $f(q)$ increases very rapidly as q is increased and in the θ - $N(X)$ region of most interest a critical level l exists such that

$$f(l) \ll 1, \quad f(l+1) \gg 1 \quad (32)$$

or

$$f(l-1) \ll 1, \quad f(l) O(1), \quad f(l+1) \gg 1. \quad (33)$$

A trifling modification of the analysis of Bates and Kingston (1964) shows that if (32) is satisfied the summation in (16) should be replaced by

$$\sum_{q=1}^{\infty} \frac{1}{\mathcal{C}(q) + \mathcal{A}(q)}. \quad (34)$$

Noting that the general replacement must be consistent with (34) in circumstance (32) and must vary smoothly with $N(X)$ we rather arbitrarily assume it to be

$$\sum_{q=q_0+1}^{\infty} \frac{\mu(q)}{\mathcal{C}(q) + \mathcal{A}(q)} \quad (35)$$

with

$$\mu(q) = \frac{f(q+1)}{1+f(q+1)}. \quad (36)$$

This choice cannot cause serious error in circumstance (33); and indeed is probably quite satisfactory even when $f(q)$ is of the order of unity for several neighbouring levels.

When radiative transitions from the continuum are significant they occur mainly to discrete levels below those giving the main contribution to summation (35) so that they simply add to the recombination rate. The total recombination coefficient is thus

$$\alpha_{\text{CR}} = \left(\frac{h^2}{2\pi m k \theta} \right)^{3/2} \left\{ \sum_{q=q_0+1}^{\infty} \frac{\mu(q)}{\mathcal{C}(q) + \mathcal{A}(q)} \right\}^{-1} + \alpha_{\text{R}} \quad (37)$$

where α_{R} is the relevant radiative recombination coefficient.

2.2. Very high densities

At very high densities recombination is hindered by the low speed with which the electrons drift towards the positive ions.† Adding reciprocal rates as in (16) we take the recombination coefficient corrected for this effect to be approximately

$$\alpha = \left\{ \frac{1}{\alpha_{\text{CR}}} + \frac{1}{\alpha_{\text{L}}} \right\}^{-1} \quad (38)$$

where

$$\alpha_{\text{L}} = \frac{3\pi e^2}{4N(X)\sigma} \left(\frac{2\pi}{m k \theta} \right)^{1/2} \quad (39)$$

is the electron-ion recombination coefficient given by the formula of Langevin (1903). The high density correction is unimportant if

$$\alpha_{\text{CR}} \ll \alpha_{\text{L}}. \quad (40)$$

We see from (11) and (39) that condition (40) is equivalent to

$$N(X)^2 \ll N_{\text{P}}(X)^2 \quad (41)$$

where

$$N_{\text{P}}(X) = \frac{3k\theta}{8\sigma e^2} \left(\frac{\pi\nu M}{2m} \right)^{1/2} = 4.3 \times 10^{20} \frac{(\nu M_{\text{Ch}})^{1/2} \theta}{\sigma_{\text{A}}} \text{ cm}^{-3} \quad (42)$$

† This effect arises at densities less than the least density at which the life² of an electron in any of the levels contributing appreciably to summation (35) is comparable with the orbiting time.

with

$$\nu = \alpha_P / \alpha_{CR} \quad (43)$$

(which number is greater than unity) and with the subscript A signifying that σ is in units of a_0^2 ($2.8 \times 10^{-17} \text{ cm}^2$). It is apparent from (42) that the density has to be extremely high before any correction to α_{CR} need be made.

Pitaevskii laid down a condition for the validity of (11) which is almost the same as (41), his expression for $N_F(X)$ differing from (42) only in that it is, apart from the factor $\nu^{1/2}$, $8 \times 2^{1/2}/3 \times \pi^{1/2}$ times larger.

2.3. Energy-change rate coefficient

The velocity of the colliding atom relative to the centre of mass of the companion electron and ion is initially

$$\mathbf{U} = \mathbf{V}_{a1} - \frac{m}{m+M} \mathbf{V}_{e1}. \quad (44)$$

Since m is much smaller than M we have from (8) that if the electron may be treated as a classical particle the change in its energy of motion relative to the ion is very closely

$$\Delta = -m\mathbf{U} \cdot (\mathbf{V}_{e1} - \mathbf{V}_{e2}) - \delta \quad (45)$$

with

$$\delta = \frac{2m^2}{M} \mathbf{V}_{e1} \cdot (\mathbf{V}_{e1} - \mathbf{V}_{e2}). \quad (46)$$

In the first instance we shall neglect δ as did Pitaevskii (1962) and take

$$\Delta_a = -m\mathbf{U} \cdot (\mathbf{V}_{e1} - \mathbf{V}_{e2}). \quad (47)$$

but we shall see later that this neglect is not fully justified and shall introduce an approximate correction. We shall also disregard the slight difference between the magnitudes of the vectors $\mathbf{V}_{e1}$ and $\mathbf{V}_{e2}$ and put

$$V_{e1} = V_{e2} = V. \quad (48)$$

Assuming that the scattering of the electron by the atom is spherically symmetrical and that the cross section σ is independent of the velocity, we find that the two-body rate coefficient describing collisions in which the energy E is changed by between Δ_a and $\Delta_a + d\Delta_a$ is $K(E, \Delta_a) d\Delta_a$ with

$$K(E, \Delta_a) = \int_{v_c}^{\infty} \mathcal{J}(u) f(u) du \quad (49)$$

$$\mathcal{J}(u) = \begin{cases} \sigma \int_{v_0}^{v_c} \left\{ 2 - \frac{|\Delta_a|}{mvu} \right\} g(v) dv & v_0 < v_c \\ 0 & v_0 > v_c \end{cases} \quad (50)$$

where $f(u)$ and $g(v)$ are the normalized distribution functions; where

$$v_0 = |\Delta_a|/2mu \quad (51)$$

is the least value of v consistent with (47); where v_c is a cut-off value defined by

$$\frac{1}{2}mv_c^2 = E + \frac{e^2}{r_c} \quad (52)$$

in which r_c is the closest the electron can be to the ion and yet be considered as colliding

with the atom; and where

$$u_c = |\Delta_a|/2mv_c. \quad (53)$$

The interaction between the ions and the atoms is ignored except in so far as it brings about thermal equilibrium between the two species with

$$f(u) = \left(\frac{M^3}{4\pi k^3 \theta^3} \right)^{1/2} u^2 \exp\left(-\frac{Mu^2}{4k\theta}\right). \quad (54)$$

Neglecting the departure of the potential from its asymptotic Coulomb form and assuming that all elements of phase space corresponding to (negative) energy E are equally populated, we have that

$$g(v) = \frac{32v^2|(2/m)E|^{5/2}}{\pi\{v^2 + (2/m)|E|\}^4}. \quad (55)$$

On substituting from (54) and (55) in (49) and (50) and carrying out some simple analysis we obtain

$$K(E, \Delta_a) = \frac{4\sigma M^{1/2}}{3m(k\theta)^{1/2}\pi^{3/2}} \int_{x_c}^{\infty} \left\{ \frac{x^{3/2}}{(1+x)^3} - \frac{x_c^3 x^{-3/2}}{(1+x_c)^3} \right\} e^{(-\gamma x)} dx \quad (56)$$

where

$$\gamma = \frac{M\Delta_a^2}{32mk\theta|E|} \quad (57)$$

and

$$x_c = 2|E|/mv_c^2. \quad (58)$$

Before proceeding it may be noted in parenthesis that if we take r_c to be zero then

$$\left(\frac{\partial \Delta^2}{\partial t} \right)_E = \frac{4\sigma M^{1/2}}{3m(k\theta)^{1/2}\pi^{3/2}} \int_0^{\infty} \int_0^{\infty} \frac{x^{3/2} e^{(-\gamma x)}}{(1+x)^3} \Delta^2 dx d\Delta; \quad (59)$$

and when the integrations on the right are carried out this reduces to formula (10) which Pitaevskii derived without introducing the rate coefficient.

The rate coefficient $\kappa(p, q)$ for collisions causing transitions from level p to level q is required. An approximation to it, which we shall denote by $\kappa_a(p, q)$ may be obtained from $K(E, \Delta)$ $d\Delta$ by putting

$$E = -I/p^2 \quad (60)$$

$$\Delta_a = I \left\{ \frac{1}{p^2} - \frac{1}{q^2} \right\} \quad (61)$$

$$d\Delta_a = 2I/q^3 \quad (62)$$

where I , as before, is the ionization potential of hydrogen. This procedure is only justified if p and q are very large but we shall tentatively use it without restriction.

We shall now consider the assumption that the energy δ defined in (46) is so small that it may be neglected. Though the assumption is plausible it is actually incorrect as is evident from the fact that it leads to serious violation of the principle of detailed balancing. The rate coefficient obtained from $\kappa_a(q, p)$ by the application of this principle is

$$\kappa_a'(p, q) = \frac{q^2}{p^2} \kappa_a(q, p) e^{(\epsilon(q) - \epsilon(p))} \quad (63)$$

and the ratio

$$\rho(p, q) = \kappa_a'(p, q)/\kappa_a(p, q) \quad (\text{in which } p > q) \quad (64)$$

may greatly exceed unity: for example, the He calculations described in the next section show that if θ is 250 °K then

$$\rho(12, 11) = 2.2, \quad \rho(7, 6) = 1.0 \times 10^2, \quad \rho(4, 3) = 5.6 \times 10^3. \quad (65)$$

The reason for δ being more important than might have been expected from the minuteness of m/M is that a considerable contribution to integral (49) comes from the part of vu space where v/u is very large. This arises because $g(v)$ is, apart from the cut-off, a much less rapidly decreasing function of v than $f(u)$ is of u throughout most of the region where

$$vu > |\Delta_a|/2m. \quad (66)$$

The use of the full expression for Δ in the calculation of the rate coefficient is difficult. However, there are two simple approximate methods of taking the awkward additional term δ into account.

Since

$$\Delta_a = \Delta + \delta \quad (67)$$

we see from the form of (46) that (61) underestimates the average value of $|\Delta_a|$ in an upward transition and that it overestimates the average value of $|\Delta_a|$ in the inverse downward transition by about the same amount. In view of this and of the nature of the dependence of the rate coefficient on $|\Delta_a|$ it is a good approximation to take

$$\kappa(p, q) = \{\kappa_a(p, q)\kappa_a'(p, q)\}^{1/2}. \quad (68)$$

Alternatively, bearing in mind that the neglect of δ is most serious when v is large and u is small, we see that the error due to this neglect may be reduced by determining the value of u corresponding to the cut-off v_c from (45) instead of from (47). We find that (53) should be replaced by

$$u_c = \frac{|\Delta_a|}{2mv_c} \left\{ 1 - \frac{\Delta_a}{2mv_c^2} + \frac{4m^2v_c^2}{M\Delta_a} \right\} \quad (69)$$

approximately, and hence that (58) should be replaced by

$$\kappa_c = \frac{2|E|}{mv_c^2} \left\{ 1 - \frac{\Delta_a}{mv_c^2} + \frac{8m^2v_c^2}{M\Delta_a} \right\}. \quad (70)$$

We preferred to use formula (68) as (70) undercorrects in the case of moderate quantum numbers and the lower temperatures: for He it undercorrects appreciably† when p and q are near 10 and θ is less than 1000 °K.

3. Results and discussion

We have treated recombination in helium



in some detail.

† The ratio of some of the values of $\kappa(p, p-1)$ obtained from formula (70) to the corresponding values obtained from (68) are as low as 0.35 at 125 °K, as 0.55 at 250 °K, and as 0.70 at 500 °K. However, the ratios of the recombination coefficients based on the two sets of values lie between 0.75 and unity.

In computing the energy change rate coefficient $\kappa(p, q)$ from (68) we took the cross section σ appearing in (56) to be $20a_0^2$ (cf. Moiseiwitsch 1962). There is little uncertainty in this choice. The position regarding r_0 of (52) is much less satisfactory. We rather arbitrarily identified it with the distance of closest approach of a slow He atom and He^+ ion. This distance is $0.76 \text{ \AA}$ (cf. Mason and Vanderslice 1962). Figure 1 shows the values obtained for $\kappa(q, q-1)$ which is the most important set of rate coefficients. Except in the region of high quantum numbers the possible error in a $(\kappa(q, q-1), q)$ curve is considerable if viewed as a vertical displacement. As we shall see later this is probably not very important as far as α is concerned.

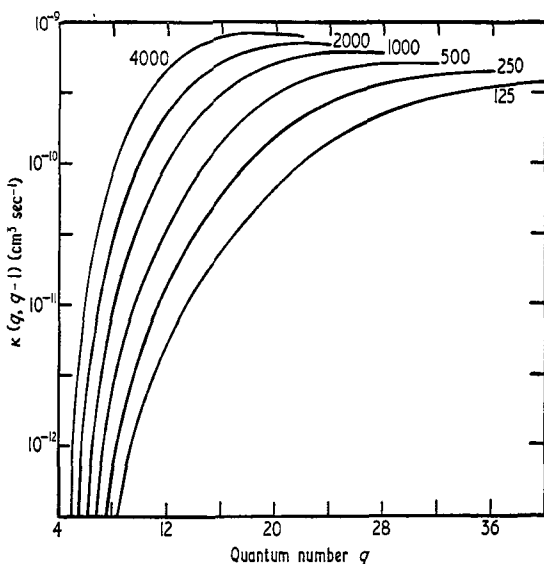


Figure 1. Energy-change rate coefficient $\kappa(q, q-1)$ in helium. The temperature θ ($^{\circ}\text{K}$) is indicated on each curve.

The radiative terms involved in (37) are known with sufficient accuracy. It is a good approximation to assume that the spontaneous transition probabilities between excited levels of helium and the radiative recombination coefficients into these levels are hydrogenic. For the former we use the values given by Baker and Menzel (1938) and by Green, Rush and Chandler (1957); and for the latter the values quoted by Bates and Dalgarno (1962). We ignored direct radiative recombination to the ground level, just as we ignored discrete radiative transitions to this level, it being supposed that the mean free path of the emitted photons is much less than the linear dimensions of the volume of the gas. In all cases q_0 is 2 or greater, and assuming it to be 2 does not cause appreciable error.

Finally, we calculated the recombination coefficient α of (38) for temperature θ from 125°K to 4000°K and for number densities $N(\text{He})$ from 10^{17} cm^{-3} to 10^{21} cm^{-3} . The reciprocal of α_L of (39) is negligible compared with the reciprocal of α_{CR} of (37) even at the lowest θ and highest $N(\text{He})$ in the range covered.

Figure 2 shows (α, θ) curves (with $N(\text{He})$ fixed). The corresponding (α_T, θ) and (α_P, θ) curves, obtained from formula (5) of Thomson (1924) and formula (11) of Pitaevskii (1962), are included for comparison.†

† Pitaevskii, however, placed the limit of applicability of his formula at about 1500°K for helium.

When θ is high α is very small so that process (71) is of little interest; but when θ is low α may greatly exceed even the rate coefficient α_d of dissociated recombination



The value of α_d is still poorly determined. Oskam and Mittelstadt (1963) have summarized the laboratory studies which have been made. According to their own measurements α_d is less than $4 \times 10^{-9} \text{ cm}^3 \text{ sec}^{-1}$ when θ is about 300 °K. There is some evidence that α_d varies as $\theta^{-3/2}$ for $300 \text{ °K} < \theta < 1500 \text{ °K}$ (Chen, Leiby and Goldstein 1961).

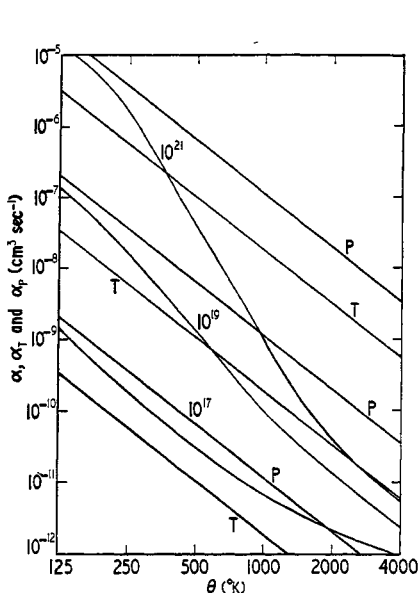


Figure 2. Recombination coefficients α , α_T and α_P for $\text{He}^+ + e + \text{He} \rightarrow 2\text{He}$ as functions of temperature. The unmarked curves give the value α obtained in the present calculations: the straight lines marked T and P give the values α_T and α_P of Thomson and of Pitaevskii respectively. The helium atom number density $N(\text{He})$ in cm^{-3} is indicated on each set of three.

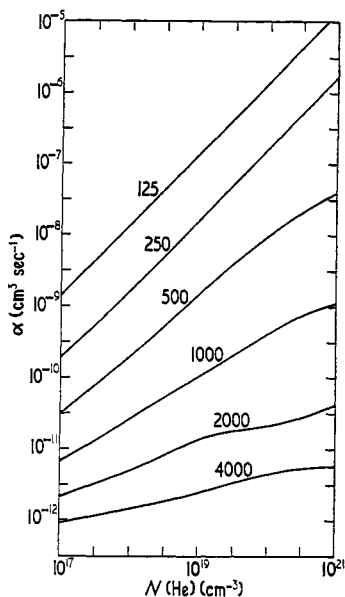


Figure 3. Recombination coefficient α for $\text{He}^+ + e + \text{He} \rightarrow 2\text{He}$ as function of the helium atom number density. The temperature θ (°K) is indicated on each curve.

Though α tends towards α_P as θ tends towards zero, the approach is very slow and it lies closer to α_T over a considerable range. As θ is increased towards 4000 °K α falls off more rapidly than do α_T and α_P if $N(\text{He})$ is of the order of 10^{19} cm^{-3} or greater but it falls off less rapidly if $N(\text{He})$ is lower. The more rapid fall-off in the former case occurs because formula (37) takes into account the discrete nature of the negative energy spectrum. This has a very marked effect when θ is high, the recombination rate being then controlled by transitions between widely spaced levels (cf. table 1). The radiation terms in formula (37) are responsible for the less rapid fall-off in the latter case.

These radiation terms also cause the departure from linearity of some of the $(\alpha, N(\text{He}))$ curves (with θ fixed) displayed in figure 3. The rather complicated pattern of the curves for the higher θ is real.

Table 1. Recombination mechanism

θ	$N(\text{He}) = 10^{17}$			$N(\text{He}) = 10^{19}$			$N(\text{He}) = 10^{21}$		
	14	19	30	14	19	30	14	19	30
125	4.4^{-1} 9.5^{-1}	1.0^0	5.1^{-1} 1.0^0	4.8^{-1} 1.0^0	1.0^0	5.1^{-1} 1.0^0	4.8^{-1} 1.0^0	1.0^0	5.1^{-1} 1.0^0
250	6.4^{-1} 7.8^{-1}	9.6^{-1}	5.2^{-1} 1.0^0	6.4^{-1} 9.9^{-1}	1.0^0	4.9^{-1} 1.0^0	6.4^{-1} 1.0^0	1.0^0	4.9^{-1} 1.0^0
500	2.2^{-1} 1.9^{-1}	10 7.8^{-1}	14 4.6^{-1} 9.9^{-1}	6 1.6^{-1} 7.9^{-3}	7 6.9^{-1}	10 5.6^{-1} 1.0^0	5 1.1^{-2} 3.3^{-3}	6 4.4^{-1}	7 1.2^{-1} 1.0^0
1000	7 3.3^{-1} 1.2^{-1}	9 7.3^{-1}	11 4.9^{-1} 9.5^{-1}	5 4.6^{-2} 1.3^{-5}	6 3.0^{-1}	7 4.5^{-1} 9.4^{-1}	4 2.5^{-5} 4.3^{-26}	5 1.3^{-3}	6 2.3^{-1} 9.8^{-1}
2000	6 3.1^{-1} 5.0^{-2}	7 3.8^{-1}	9 5.7^{-1} 9.1^{-1}	4 1.8^{-3} 2.9^{-14}	5 1.3^{-2}	6 7.2^{-1} 8.4^{-1}	4 1.5^{-1} 2.9^{-12}	5 5.7^{-1}	6 1.7^{-1} 1.0^0
4000	5 1.9^{-1} 6.8^{-3}	6 2.3^{-1}	8 5.1^{-1} 9.0^{-1}	4 6.5^{-2} 3.9^{-7}	5 4.1^{-1}	6 2.9^{-1} 9.6^{-1}	3 4.1^{-6} 4.1^{-61}	4 3.9^{-5}	5 5.3^{-2} 1.0^0

The entries for each temperature θ °K and density $N(\text{He})$ cm $^{-3}$ are arranged in the form

$$\begin{array}{ccc} q_{\frac{1}{2}x}^- & q_x & q_{\frac{1}{2}x}^+ \\ r_{\frac{1}{2}x}^- & & r_{\frac{1}{2}x}^+ \\ \mu(q_{\frac{1}{2}x}^- - 1) & \mu(q_x - 1) & \mu(q_{\frac{1}{2}x}^+ - 1) \end{array}$$

the notation being as defined in the text. The indices give the power of 10 by which the printed numbers in the second and third rows of each set must be multiplied.

Table 1 provides information relating to the recombination mechanism. It gives (i) the quantum number q_x at which the term

$$t(q) = \frac{\mu(q)}{\mathcal{C}(q) + \mathcal{A}(q)} \quad (73)$$

of the summation in (37) is greatest; (ii) the values $q_{\frac{1}{2}x}^-$ and $q_{\frac{1}{2}x}^+$ of q below and above q_x at which $t(q_{\frac{1}{2}x}^-)$ and $t(q_{\frac{1}{2}x}^+)$ are closest to $\frac{1}{2}t(q_x)$ so that $q_{\frac{1}{2}x}^+ - q_{\frac{1}{2}x}^-$ is a measure of how many terms effectively determine the recombination coefficient; (iii) the actual ratios

$$r_{\frac{1}{2}x}^- = t(q_{\frac{1}{2}x}^-)/t(q_x), \quad r_{\frac{1}{2}x}^+ = t(q_{\frac{1}{2}x}^+)/t(q_x) \quad (74)$$

which may be different from $\frac{1}{2}$ and which must be borne in mind when considering the ratio just mentioned; and (iv) the factors $\mu(q_{\frac{1}{2}x}^- - 1)$, $\mu(q_x - 1)$ and $\mu(q_{\frac{1}{2}x}^+ - 1)$ where $\mu(q)$ is as defined in (36) and thus indicates the importance of the radiation term in $t(q)$ being less than $\frac{1}{2}$ if this term exceeds the collisional term.

It may be seen that q_x and $q_{\frac{1}{2}x}^+ - q_{\frac{1}{2}x}^-$ are quite large when θ is low. However, both decrease sharply as θ is raised (especially if $N(\text{He})$ is high): for example, at 1000 °K and 10^{19} cm $^{-3}$ q_x is only 6 and transitions from this level and from level 7 together

provide almost all the resistance to recombination. In connection with Thomson's theory it is of interest to note that the energy of level q_x ranges from just over 1 to almost 9 in units of $k\theta$.

The radiation terms are most important when θ is high. Their presence helps to limit the possible inaccuracy in α due to the uncertainties in the collisional terms. Since q_x is not large it might be feared that this inaccuracy is very great. To examine the position we carried out some calculations with r_c of (52) taken to be 0.5 Å and 1.0 Å instead of 0.76 Å. Table 2 shows the factors $k(0.76 \rightarrow 0.5)$ and $k(0.76 \rightarrow 1.0)$, by which the original $\kappa(q_x, q_x - 1)$ is multiplied as a result of the changes together with the corresponding factors $a(0.76 \rightarrow 0.5)$ and $a(0.76 \rightarrow 1.0)$ by which the original α is multiplied. Their pattern is rather complicated. The implication appears to be that the curves in figures 2 and 3 should not be seriously in error.

Table 2. Reliability of Results

θ	$N(\text{He}) = 10^{17}$		$N(\text{He}) = 10^{19}$		$N(\text{He}) = 10^{21}$	
250	1.0°	9.9 ⁻¹	1.0°	9.9 ⁻¹	1.0°	9.9 ⁻¹
	1.0°	9.9 ⁻¹	1.0°	9.7 ⁻¹	1.0°	8.6 ⁻¹
1000	1.0°	9.7 ⁻¹	3.2°	3.1 ⁻¹	9.8 ¹	1.5 ⁻²
	1.0°	9.9 ⁻¹	1.3°	6.3 ⁻¹	1.2°	7.0 ⁻¹
4000	1.2°	8.0 ⁻¹	2.7°	3.6 ⁻¹	9.6 ²	3.7 ⁻⁴
	1.0°	9.9 ⁻¹	1.3°	8.4 ⁻¹	1.0°	9.4 ⁻¹

The entries for each temperature θ °K and density $N(\text{He})$ cm⁻³ are arranged in the form

$$\begin{array}{ll} k(0.76 \rightarrow 0.5), & k(0.76 \rightarrow 1.0) \\ a(0.76 \rightarrow 0.5), & a(0.76 \rightarrow 1.0) \end{array}$$

the notation being as described in the text. The indices give the power of 10 by which the printed number must be multiplied.

Without detailed calculations little can usefully be said about recombination in other cases. According to the theories of Thomson and Pitaevskii α is proportional to $\sigma\bar{M}^{-1}$ where $\bar{M}$ is the reduced mass of the ion and atom; and if θ is sufficiently low this behaviour is certainly followed. However, using formula (56) it is found that in the important region near N.T.P. taking α to be proportional to $\sigma\bar{M}^{-2}$ is a better approximation at least if $\bar{M}$ is greater than 2 on the chemical scale.

The following are the values of α obtained for some of the alkali ions in helium gas with θ at 250 °K and $N(\text{He})$ at 10^{19} cm⁻³: Li⁺, 1.1×10^{-8} cm³ sec⁻¹; Na⁺, 6.9×10^{-9} cm³ sec⁻¹; K⁺, 6.0×10^{-9} cm³ sec⁻¹.

In general, of course, it is not permissible to treat σ as a constant; it is necessary to take into account the dependence on the velocity. If the third body is a molecule allowance must be made for the internal degrees of freedom. Doing this is complicated by the fact that the critical transitions of the electron are between discrete levels.

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